

Praveen De Silva

Computer Science and Engineering Undergraduate, University of Moratuwa

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🌐 LinkedIn | 🌐 Portfolio | 🏠 HackerRank

PROFILE

Computer Science and Engineering undergraduate at the University of Moratuwa specializing in Data Science and Engineering, with practical experience in Python, SQL, data analysis and machine learning. Experienced in preparing and analyzing real-world datasets through research and data-driven projects, with a strong interest in data engineering and AI-driven analytics.

AWARDS

Dean's List	2024 / 25
1st semester GPA: 4.00, 2nd semester GPA: 3.82 University of Moratuwa	
Global Rank 68, IEEEExtreme 19.0	2025
Among 19,000+ participations IEEE	
9th Place, MiniXtreme Problem Solving Competition	2025
Top 10 of 200+ in CSE'23 batch University of Moratuwa	
Finalist	2025
SLIOT Challenge 2025 University of Moratuwa	
CodeArena 2025 KDU Sri Lanka	
Semi-Finalist	2025 / 26
SLIOT Challenge 2026 University of Moratuwa	
Mahapola Scholarship recipient	2022
Physical Science stream Kingswood College	

PUBLICATIONS

Pre-Match ODI Win Probability Prediction: Role-Based Feature Engineering and Training Data Analysis for Sri Lanka Cricket	2026
<i>Proceedings of the Undergraduate Research Symposium 2026, Dept. of Computer Engineering, University of Jaffna (ISBN: 978-624-6150-99-0)</i>	
<i>Python, scikit-learn, XGBoost, pandas, Feature Engineering, Cricsheet</i> · 🐙 Code · 🌐 Post	
Developed and evaluated five machine-learning classifiers using historical cricket data for pre-match win-probability prediction. Engineered role-based features that improved performance over aggregate baselines by up to 20.86 percentage points. Achieved 71.30% accuracy and 0.78 ROC-AUC with Random Forest; selected as 1 of 6 podium presenters at the Undergraduate Research Symposium 2026.	

RESEARCH

Physics-Informed, GAN-Augmented Graph Neural Networks for Multi-Horizon Dengue Forecasting	2026
<i>Ongoing research, University of Moratuwa</i>	
<i>PyTorch Geometric, Spatio-Temporal GNNs (GCN/GAT), Physics-Informed Learning, GANs</i>	
Building a spatio-temporal GNN to forecast weekly, district-level dengue incidence across Sri Lanka, enhanced with an epidemic-mechanistic (SEIR-SEI) physics-informed loss, conditional GAN-based augmentation to data scarcity, and a learned adaptive graph capturing unobserved inter-district transmission.	
Hidden Imbalance: Machine Learning Evidence of Underemployment in Sri Lanka	2026
<i>Manuscript in preparation</i>	
<i>Machine Learning, Econometrics, XGBoost, SHAP, ARDL, Python</i>	
Investigated underemployment trends in Sri Lanka using macroeconomic and labour market indicators.	

PROJECTS

B-Trust MBMS	Group
<i>PostgreSQL, Express.js, React, Node.js, TypeScript</i> · 🐙 Code	
Designed and implemented a PostgreSQL-based micro-banking system with automated interest calculations, ACID-compliant transactions and role-based access control. Developed REST APIs and optimized indexed database queries.	

EcoEYE - SLIOT'26 (Semi-Finalist)

Group

Python, OpenCV, YOLO-v8-n, Raspberry Pi, ESP32, Arduino, React · [Code](#) · [Post](#)

Developed an IoT energy-management system to detect occupancy from CCTV feeds and trigger zone-based control of building appliances.

RPAL Interpreter (Completed)

Group

C++, Python, HTML/CSS, Render · [Code](#) · [Post](#) · [Demo](#)

Implemented a complete 6-stage interpreter pipeline for RPAL.

Draftly - Lawyer-in-the-Loop Legal Workflow Platform (In Progress)

Group

Google Document AI, RAG, NLP, Full-Stack Web, local llm

Legal workflow platform for Sri Lankan practice that turns unstructured client documents into a verified, evidence-linked matter record with deterministic checks, grounded legal search, and reviewable drafting.

Supervised by Dr. Nisansa de Silva.

Flower Exchange - Order Matching Engine

Individual

C++17, OOP / SOLID, REST API, Web UI · [Code](#)

Trading exchange with a price-time-priority order matching engine and both CLI and web interfaces, built on SOLID design principles. Developed in a selective LSEG C++ workshop offered to 25 students (after applied) from the batch.

CoupleCore (In Progress)

Individual

Flutter, Firebase, Cloud Functions, LLM Integration · [Code](#)

Relationship companion app whose core innovation is an AI problem-solving tool that learns from each couple's own history of past issues and resolutions to generate tailored, context-aware guidance.

EDUCATION

University of Moratuwa

2024 - 2028

BSc Engineering Honours in Computer Science & Engineering

Specializing in Data Science and Engineering | CGPA: 3.56

ESOFT Metro Campus

2022 - 2023

Diploma in Information Technology, Diploma in English | Overall Grades: 02 Distinctions

Kingswood College, Kandy

2008 - 2022

GCE A/L: 3As (Physical Science) | IR: 152 | Z-score: 2.5895

GCE O/L: 9As (including Tamil)

ACTIVITIES & EXPERIENCES

Batch Rep

DiTec-195, Esoft Metro Campus Kandy

Web Developer, Media Committee

Build and maintain the official website, IEEE WIE Student Branch, University of Moratuwa

Violinist and Member

Classical Music Society, University of Moratuwa, Kingswood College Kandy

1st Place, All Island Music Competition (School Category - Orchestra, Violinist)

Volunteer

Susanaka Sansada (2023 - Present)

CSE '23 Careers Day, University of Moratuwa

TECHNICAL SKILLS

Languages

Python, SQL, C++, C, Java, TypeScript, Dart

Frameworks / Libraries

React, Node.js, FastAPI, Flutter, PyTorch Geometric, scikit-learn, XGBoost

Database

PostgreSQL, MongoDB, Firebase

Tools

Google Cloud, Docker, Git

REFERENCES

Prof. Dulani Meedeniya

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Dr. Adeesha Wijayasiri

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